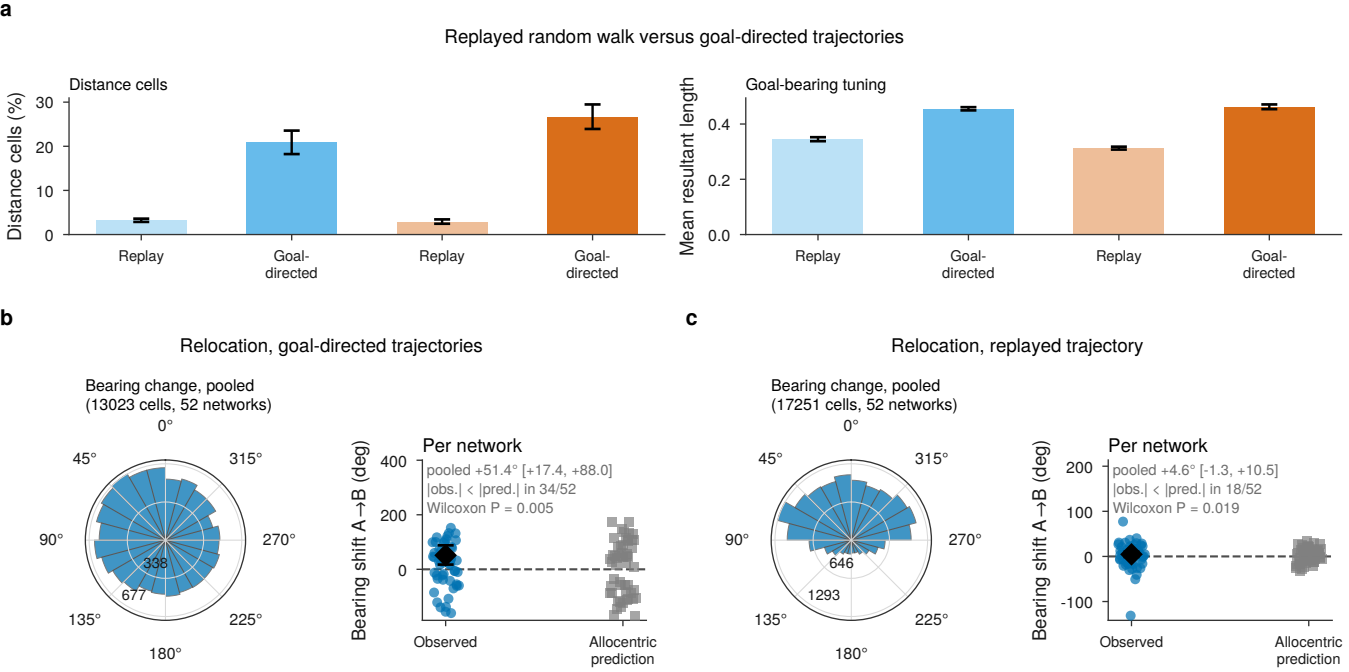

a
Replayed random walk versus goal-directed trajectories
Distance cells
Distance cells (%)
30
20
10
0
Replay
Goal-directed
Replay
Goal-directed
Goal-bearing tuning
Mean resultant length
0.4
0.2
0.0
Replay
Goal-directed
Replay
Goal-directed
b
Relocation, goal-directed trajectories
Bearing change, pooled
(13023 cells, 52 networks)
0°
45°
315°
90°
270°
135°
225°
180°
338
677
Per network
pooled +51.4° [+17.4, +88.0]
|obs.| < |pred.| in 34/52
Wilcoxon P = 0.005
Bearing shift A→B (deg)
400
200
0
Observed
Allocentric prediction
c
Relocation, replayed trajectory
Bearing change, pooled
(17251 cells, 52 networks)
0°
45°
315°
90°
270°
135°
225°
180°
646
1293
Per network
pooled +4.6° [-1.3, +10.5]
|obs.| < |pred.| in 18/52
Wilcoxon P = 0.019
Bearing shift A→B (deg)
200
100
0
-100
Observed
Allocentric prediction